

Experimental text for draft integration

September 14, 2026

1 Experiments

We distinguish two questions. First, does conditioning the actual frozen policy on feasibility improve useful behavior relative to projecting its samples? Second, does a finite implementation of our refinement mechanism recover those benefits and approximate the conditional distribution? These questions need separate controls: improved return alone does not establish conditional sampling.

Controlled protocol. Let $p_{\text{sampler}}(A \mid s)$ denote the distribution produced by the frozen 100-step DDPM, including normalization and physical action clipping. Our reference returns the first feasible independent proposal, without ranking accepted proposals by reward. It samples the feasible conditional law given budget success. All proposals and budget failures are retained. Projection minimizes squared Euclidean distance over the same complete action chunk and feasible set. Every method receives the same observation and constraints, executes the same chunk duration, and then uses the same deterministic public RL controller. We measure unchanged native return, feasibility, failures, boundary concentration, action-kernel discrepancy, and cost.

We evaluate Pendulum with a 0.6 s chunk and 10 s total rollout, using mirrored initial torque envelopes $[-2, 0]^{12}$ and $[0, 2]^{12}$. Walker2d and HalfCheetah use 0.096 s and 0.4 s chunks respectively, followed by continuation through 4 s total. Their feasible sets intersect the physical action box with a whole-chunk RMS actuator-command budget. These convex constraints admit global projection: coordinate clipping for Pendulum and the scalar KKT solution for a box intersected with a Euclidean ball. The RMS envelope is an effort proxy applied to the initial chunk; it does not certify subsequent state safety or measure mechanical energy.

Pendulum uses 64 held-out states and 8 replicates per state. Each MuJoCo task uses 32 full simulator snapshots and 4 replicates; half are reset states and half follow 1 s of expert warmup. Observations, model weights, constraints, seeds, horizons, and analysis rules were frozen before evaluation. Rejection budgets are 64 proposals per Pendulum request and 128 per MuJoCo request. State-cluster bootstrap intervals account for repeated samples from one state. The new Pendulum baseline comparisons use a four-comparison correction; MuJoCo’s two primary contrasts at four settings use an eight-comparison correction. These intervals condition on the trained policies and do not estimate variation across independent training runs.

Policies and finite samplers. The Pendulum DDPM imitates complete episodes from a public TD3 controller and its reflected counterpart. For each MuJoCo task, we acquired released Farama-Minari SAC expert and medium controllers and collected 128 complete episodes from 64 paired reset groups. Groups were split 48/8/8 before overlapping action chunks were constructed, and observation normalization used training episodes only. Initial diffusion fits failed the development quality gate. A single bounded refit shortened Walker2d’s chunk and increased early-episode sampling; both resulting policies passed the gate before safety comparisons. All initial fits and failed outcomes are retained.

Table 1: Mean native return after one constrained chunk and identical continuation. Q: first-feasible rejection; P: exact projection; R: finite refinement; B+P: paper-derived BayesFP with 32 particles and exact terminal repair. Pendulum uses 64 states with 8 replicates and 10 s evaluation; MuJoCo uses 32 states with 4 replicates and 4 s evaluation.

Setting	Q	P	R	B+P
Pendulum negative	-367.01	-423.38	-366.35	-366.72
Pendulum positive	-377.90	-425.67	-378.02	-376.82
Walker2d 0.72	3302.52	3320.09	3336.20	3314.86
Walker2d 0.75	3314.99	3321.23	3316.10	3320.19
HalfCheetah 0.7	442.08	440.89	461.22	447.21
HalfCheetah 0.72	454.09	454.81	469.15	438.54

Table 2: Paired return differences with state-cluster bootstrap intervals. Q-P: 95% (Pendulum, descriptive), 99.375% (MuJoCo). R-(B+P): 98.75% (Pendulum), 99.375% (MuJoCo), adjusting their respective four- and eight-comparison families. Positive favors the first method. These are approximate bootstrap intervals, not finite-sample guarantees.

Setting	Q-P [interval]	R-(B+P) [interval]
Pendulum negative	+56.37 [+51.11, +61.54]	+0.38 [-0.99, +2.06]
Pendulum positive	+47.77 [+43.08, +52.37]	-1.20 [-2.44, +0.21]
Walker2d 0.72	-17.57 [-29.68, -5.94]	+21.35 [+11.06, +31.82]
Walker2d 0.75	-6.24 [-12.45, -0.41]	-4.09 [-95.29, +32.03]
HalfCheetah 0.7	+1.19 [-30.11, +35.40]	+14.01 [-21.90, +49.63]
HalfCheetah 0.72	-0.73 [-8.99, +9.38]	+30.61 [+5.44, +57.42]

Our finite sampler uses $\gamma = 1/2$ and an annealed refinement schedule with 256, 256, and 512 updates at DDPM indices 20, 5, and 0. The VP epsilon network is converted to a VE score at each level. Within each level, $\alpha_k = 0.15\sigma^2/(1 + k/64)^{0.6}$. This is a finite annealed variant, not a test of all relaxation parameters or of asymptotic convergence.

We additionally compare a paper-derived DDPM reimplementation of BayesFP (Sirigiri et al., 2026), using 32 particles and a safety-only cost. Guidance strength is selected by development discrepancy from rejection, without optimizing task reward. We report both raw output and output followed by the identical exact projection. The latter is the hard-feasible comparison. Pendulum also includes 12 particles, approximately matching our score count. This is not a reproduction using official author code or author benchmarks.

When conditioning helps. Pendulum provides the clearest mechanism: projecting the excluded swing direction concentrates samples near zero torque, delaying useful motion. Conditioning improves native return by 56.37 and 47.77 points in the two mirrored cases; finite refinement improves it by 57.04 and 47.65 points. Projection places 64.6% and 65.4% of outputs on the added zero-torque boundary, compared with zero observed added-boundary outputs for rejection and refinement. All methods eventually stabilize in this evaluation; the improvement is in continuous return and stabilization time, not a higher completion count. Each signed constraint predominantly retains one useful direction, so this experiment does not establish preservation of weights among multiple surviving safe modes.

The effect is task dependent. On Walker2d, conditioning slightly reduces return relative to projection at both effort caps. On HalfCheetah the conditioning-projection differences are inconclusive. The earlier released Push-T policy experiment also failed to produce a useful conditioning gain: at its stricter reported cap the native coverage difference favored projection. Thus boundary concentration alone does not establish a meaningful task loss.

Comparison with BayesFP and approximation limits. On Pendulum, refinement has no useful return advantage over repaired BayesFP. For HalfCheetah at RMS cap 0.72,

refinement exceeds repaired BayesFP by 30.61 points, with an adjusted interval [5.44, 57.42]. Its point estimate exceeds the preregistered 26.19-point useful-effect threshold, while the interval still allows smaller effects. Conditioning does not outperform projection in this setting, and refinement’s advantage over ordinary projection is inconclusive. This result therefore supports a finite-sampler comparison, not a causal claim that more accurate conditioning produced the gain. Walker2d’s positive refinement–BayesFP difference at the stricter cap is below its useful-effect threshold; one refinement rollout at the milder cap terminates unhealthy.

Raw 32-particle BayesFP violates the chunk constraint in 13.9–18.0% of Pendulum outputs and 18.0–57.8% of MuJoCo outputs. Rejection, projection, refinement and repaired BayesFP have zero observed chunk violations; rejection has zero budget failures. Finite refinement remains detectably different from the rejection reference. In the HalfCheetah setting above, its action-kernel MMD is 0.157, versus 0.119 for repaired BayesFP and 0.113 for independent rejection samples. These are finite-sample diagnostics, not calibrated tests of distributional equality. Recorded batch latency favors BayesFP despite its larger score count. We therefore do not claim superior online efficiency, exact conditional sampling, or universal task-quality improvement.

Reproducibility. All experiments ran in simulation on the inspected Threadripper 9980X and RTX 5070 Ti workstation. The artifact includes frozen dependency records, public asset URLs/revisions/hashes, episode splits, all proposal banks, complete outcomes including failures, solver checks, simulator replay checks, and scripts that regenerate tables and figures. A fresh-directory rerun of HalfCheetah reproduces the saved numerical arrays exactly; this is a reproducibility check rather than independent statistical evidence.

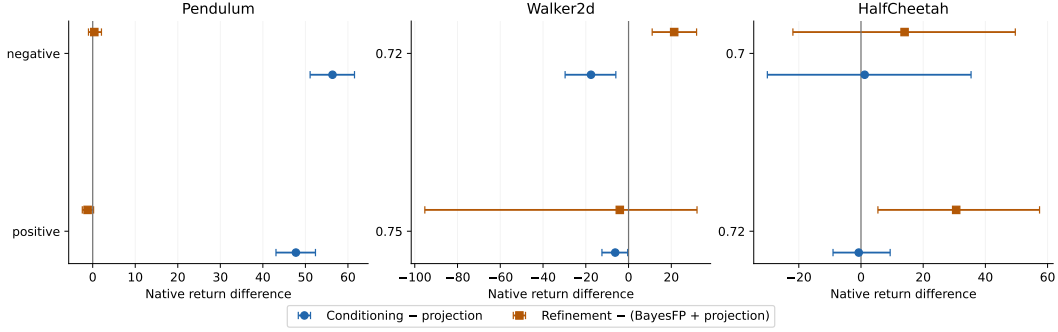


Figure 1: Target utility and finite-sampler performance are separate comparisons. The horizontal axes use each task’s native return units. Q–P intervals are 95% for Pendulum and 99.375% for MuJoCo; R–(B+P) intervals are 98.75% and 99.375%, respectively. They follow the separate predeclared comparison families described in the text.

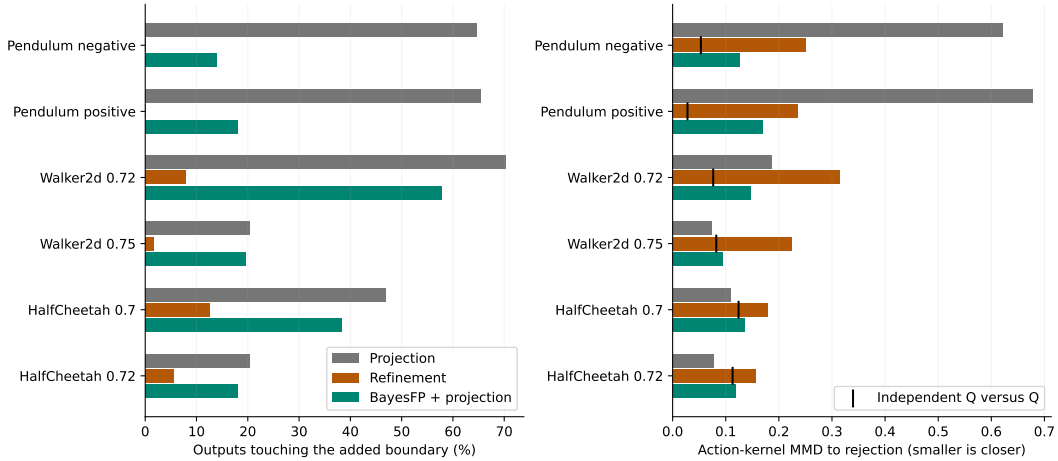


Figure 2: Boundary concentration and finite-sample action-kernel discrepancy. The independent Q-versus-Q marker indicates sampling noise, not a formal threshold. Kernel RMS bandwidth is .1 for normalized Pendulum actions and .2 for MuJoCo actions. MMD values therefore should not be compared directly across tasks. The added safety boundary is distinct from pre-existing physical action saturation.